

Spring Data JPA - Quick Example

Software Requirements:

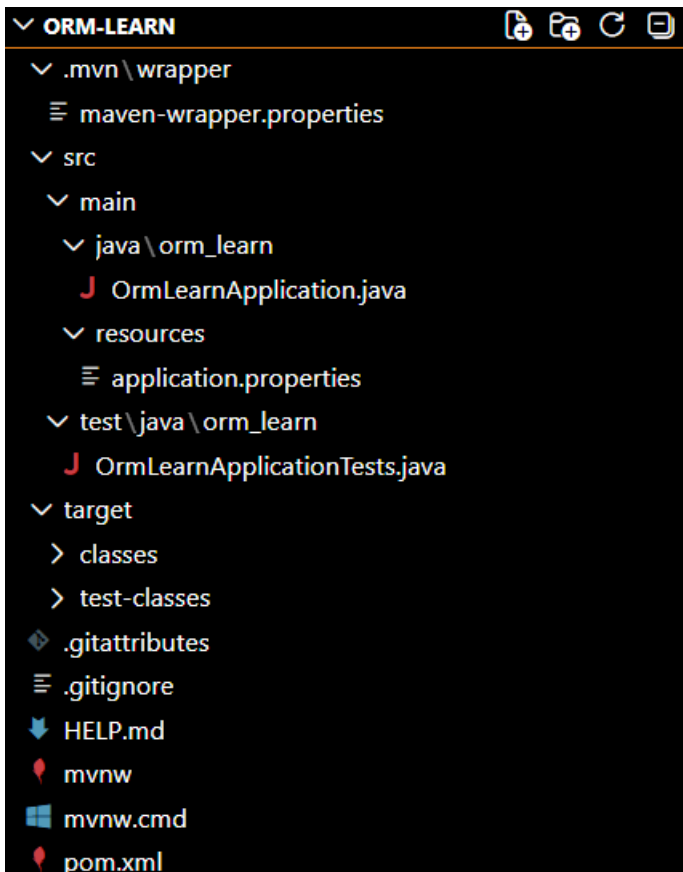
- Java JDK 17
- Spring Boot 4.1.0
- Maven
- MySQL Server 8.0
- MySQL Workbench
- Visual Studio Code

The screenshot shows the Spring Initializr web application in a browser. The interface is dark-themed and includes the following sections:

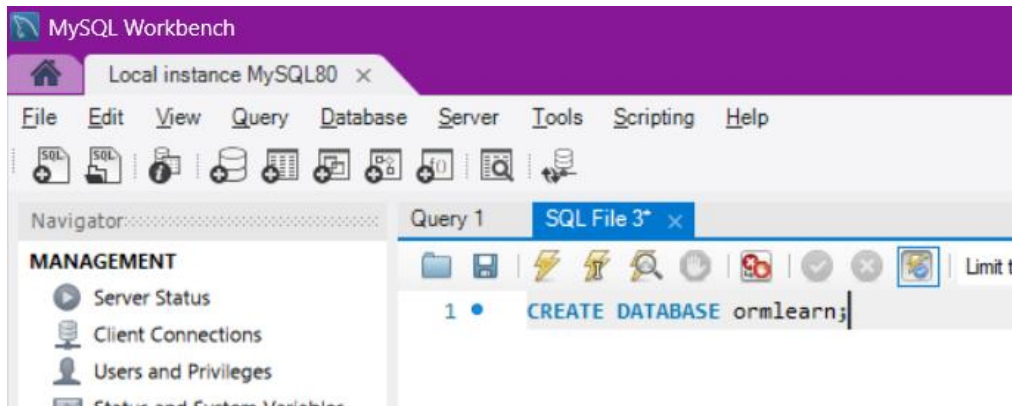
- Project:** Radio buttons for Gradle - Groovy, Gradle - Kotlin, and **Maven** (selected).
- Language:** Radio buttons for **Java** (selected), Kotlin, and Groovy.
- Spring Boot:** Radio buttons for 4.1.1 (SNAPSHOT), **4.1.0** (selected), 4.0.8 (SNAPSHOT), 4.0.7, and 3.5.16.
- Project Metadata:** Text input fields for Group (com.cognizant), Artifact (orm-learn), and Package name (orm-learn). Radio buttons for Packaging (**Jar** selected, War) and Configuration (**Properties** selected, YAML).
- Dependencies:** A list of dependencies with a button "ADD DEPENDENCIES... CTRL + B". The list includes:
 - Spring Boot DevTools** (DEVELOPER TOOLS): Provides fast application restarts, LiveReload, and configurations for enhanced development experience.
 - Spring Data JPA** (SQL): Persist data in SQL stores with Java Persistence API using Spring Data and Hibernate.
 - MySQL Driver** (SQL): MySQL JDBC driver.

At the bottom, there are buttons for "GENERATE CTRL + G", "EXPLORE CTRL + SPACE", and a menu icon "...".

File structure:



Database Creation:



application.properties:

```
# Spring Framework and application log
logging.level.org.springframework=info
logging.level.orm_learn=debug
```

```
# Hibernate logs
logging.level.org.hibernate.SQL=trace
logging.level.org.hibernate.orm.jdbc.bind=trace
```

```
# Log Pattern
logging.pattern.console=%d{dd-MM-yyyy HH:mm:ss.SSS} %-20.20thread %5p %-25.25logger{25}
%25M %4L %m%n
```

```
# Database Configuration
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
spring.datasource.username=root
spring.datasource.password=root
```

```
# Hibernate
spring.jpa.hibernate.ddl-auto=validate
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect
```

Main class:

```
package orm_learn;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class OrmLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(OrmLearnApplication.class);
```

}

Output:

[illegible]